

# Ubuntu va OS asoslari: mustaqil topshiriqlar

M0-M7 Mission Guide

Ubuntu Server 24.04 LTS / ochiq Lab v1.0.0  
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# Ubuntu and OS Foundations — mustaqil topshiriqlar: Mission Guide

Bu tarjimada amaldagi M1–M7 topshiriqlari hali aks etmagan: eski M0 endi P0 mashqidir. Lab tekshiradigan amaldagi talablar uchun OS-Linux-Control repositorysidagi yaponcha joriy nusxadan foydalaning.

Qamrov: Mission 0–7. Muhit: Ubuntu Server 24.04 LTS. Har bir talaba Missionlarni o'z sur'atida bajaradi; ular dars soatlariga birma-bir bog'lanmagan.

Bu hujjat `command`ni talaba o'zi tanlaydigan M0–M7 topshiriqlaridir. To'liq, qadamma-qadam ko'rsatma kerak bo'lsa, avval `practice.mdd`dagi P1–P6 mashqlarini bajaring.

P va M alohida resurslardan foydalanadi. P tugagani tegishli M avtomatik `PASS` bo'lishini anglatmaydi. Tavsiya etilgan tartib: `P1 → M1 → P2 → M2 → … → P6 → M6 → M7`. M0 va yakuniy M7 uchun alohida P yo'q.

## Umumiy amallar

Muhit ilk yaratilganda barcha Missionlar tugallanmagan holatda bo'ladi. Avval `reset` qilish shart emas.

### Amallarni bajaradigan user

Ubuntuga ulangandan keyingi boshqaruv `useri` — `ssm-user`. Tekshiring:

```
id -un
```

`jdu-check` va `jdu-reset`ni faqat `ssm-user` sifatida bajaring. Boshqa `user` ishlatsa, dastur `ssm-user`ga qaytish usulini ko'rsatib tugaydi.

Mashq uchun `jduops`, `jduviewer`, `jduwriter` hisoblari bor. `ssm-user`dan ulardan biriga o'tish misoli:

```
sudo su - jduops  
id
```

Asl `ssm-user`ga qayting:

```
exit  
id -un
```

Bu yerda almashtiriladigan `user`lardan faqat `ssm-user` parolsiz `sudo` ishlata oladi. Masalan, `jduwriter`dan bevosita `jduviewerga` o'tmang. Avval `exit` bilan `ssm-user`ga qayting, keyin `sudo su - jduviewer`ni bajaring.

`jduapp`, `jduweb`, `jdufinal`, `jduworker` faqat `service` uchun mo'ljallangan `user`lardir. Ularning login `shell`i `nologin`; `su -` bilan interaktiv login qilmang. Kerakli tekshiruvni `ssm-user`dan `sudo -u USER COMMAND` shaklida bitta `command` sifatida bajaring.

Hozirgi Missionni tekshiring:

```
jdu-check M1
```

O'qituvchining progress serveri sozlangan bo'lsa, natija avtomatik yuboriladi. Barcha savollar PASS bo'lishini kutmaydi.

RESULT — Ubuntudagi topshiriq natijasi. REPORT — HTTPS yuborish natijasi. Yuborish xatosi Missionning PASS/FAIL holatini o'zgartirmaydi. Anonim server IDni quyidagicha ko'ring:

```
jdu-progress id
```

Faqat local tekshiruvni bajarib, hozircha natija yubormaslik uchun:

```
jdu-check M1 --no-submit
```

Missionni boshidan boshlash zarur bo'lsagina uni reset qiling:

```
jdu-reset M1
```

resetdan darhol keyin barcha bandlar FAIL bo'ladi. Bu kutilgan holat. LabCheck holatni o'zi tuzatmaydi.

## M0 Environment and OS

Haqiqiy muhitda Ubuntu, Linux kernel, PID 1, hozirgi user va hostnamenani aniqlang. Oltita qiymatni ~/jdu-lab/m0/observation.envga yozing:

```
OS_ID=  
OS_VERSION_ID=  
KERNEL_RELEASE=  
PID1_COMM=  
USER_NAME=  
HOST_NAME=
```

Qiymatlarni ishlayotgan tizimdagi command natijasidan oling. Taxmin qilmang.

```
jdu-check M0
```

## M1 Shell, path, file, and text

### Maqsad

relative path va absolute pathni farqlash; directory va file yaratish; filedan nusxa olish; logdan kerakli satrlarni ajratish.

### Boshlang'ich holat

Materiallar ~/jdu-lab/m1/inboxda. case01 hali tayyor emas; stagingda ortiqcha .tmp file bor.

### Topshiriq

1. Quyidagi directory treeni yarating:

```
~/jdu-lab/m1/case01/  
├── config/  
├── logs/  
└── notes/
```

1. `inbox/config/app.conf`dan `case01/config/app.conf`ga nusxa oling.
2. `inbox/logs/incident.log`dan `case01/logs/incident.log`ga nusxa oling.
3. Nusxa olingan ikkala file mazmunini o'zgartirmang.
4. `case01` ichidagi barcha `.tmp` filelarni olib tashlang.
5. `case01` ostidagi obyektlarning egasi Ubuntuga kirgan boshqaruv useri `ssm-user` bo'lsin.
6. `incident.log`dagi `ERROR` bor to'liq satrlariga `notes/errors.txt`ga yozing. Satr raqami qo'shmang.
7. `incident.log`ning oxirgi besh satrini tartibini o'zgartirmay `notes/recent.txt`ga yozing.

## Tekshiruv

Oltita natija tekshiriladi: `directory`, manba filelar, `.tmp`, egasi, `errors.txt`, `recent.txt`.

```
jdu-check M1
```

Foydali commandlar: `pwd`, `ls`, `tree`, `mkdir`, `cp`, `rm`, `find`, `grep`, `tail`, `stat`.

## M2 User, group, permission, and sudo

### Maqsad

`user`, `primary group`, `supplementary group`ni farqlash; umumiy `directory`ga `group permission` va `setgid` qo'yish.

### Boshlang'ich holat

`jduops`, `jduviewer` va `ops` oldindan bor. Dastlab faqat `jduviewer` noto'g'ri ravishda `ops supplementary group`ga kirgan. `jduops` esa `ops`da emas. `/srv/jdu-share` va `README.txt`ning egasi, `group`i va `mode`i tayyor emas.

### Topshiriq

1. `jduops`ni `ops supplementary group`ga qo'shing.
2. `jduviewer`ni `ops supplementary group`idan chiqaring.
3. `/srv/jdu-share` egasi va `group`ini `root:ops` qiling.
4. `/srv/jdu-share` `mode`ini `2775` qiling; `setgid` ishlating. Hammaga yozish huquqini bermang.
5. `/srv/jdu-share/README.txt`ni `root:ops`, `mode 664` qiling.
6. `ssm-user`dan `sudo su - jduops` orqali o'ting. Umumiy `directory`da file yarating. Yangi file `group`i `ops` bo'lganini tekshiring.
7. `jduviewer` `README.txt`ni o'qiy olishini tekshiring.
8. `jduviewer` umumiy `directory`da yangi file yarata olmasligini tekshiring.

`jduops`dagi tekshiruvdan keyin `exit` bilan `ssm-user`ga qayting. So'ng `sudo su - jduviewer` orqali o'ting. `jduviewer` tekshiruvidan keyin ham `exit` qiling. Oxirida `id -un` natijasi `ssm-user` ekanini tekshirib, keyin `jdu-check M2`ni bajaring.

## Tekshiruv

Oltita natija tekshiriladi: `group` a'zoligi, `directory` egasi/`group`i, `directory mode`, namunaviy file, `group` merosi va `viewer` huquqi.

```
jdu-check M2
```

Foydali commandlar: `id`, `groups`, `getent passwd`, `getent group`, `stat`, `namei`, `chown`, `chmod`, `usermod`, `sudo -u`.

Taqiqlanadi: `chmod 777`.

## M3 Process and package

8- va 9-boblarni hamda P3 mashqining ikkala qismini tugatgach bajaring. 10-bobda `serviceni` boshlash va avtomatik boshlash batafsil ko'riladi.

### Maqsad

`service` va `processni` bog'lash; PID orqali `signal` yuborish; Ubuntu `packageini` tekshirib o'rnatish.

### Boshlang'ich holat

`jdu-m3-process1.service`, `jdu-m3-process2.service`, `jdu-m3-process3.service` odatdagi `systemd` unitlari sifatida ishlamoqda. `cmatrix` o'rnatilmagan. Boshlang'ich holatni tekshirish:

```
systemctl list-units --type=service 'jdu-m3-*'  
systemctl status jdu-m3-process1.service jdu-m3-process2.service jdu-m3-process3.service
```

### Topshiriq A: faqat process2ni to'xtatish

1. `jdu-m3-process2.service` Main PIDini toping.
2. Shu PIDdagi `processni` `ps` bilan tekshiring.
3. Faqat shu PIDga `TERM` yuboring.
4. `process2` to'xtagan, `process1` va `process3` ishlayotganini tekshiring.

Keng qamrovli `pkill`, `killall` yoki `KILL` signalini ishlatmang.

### Topshiriq B: cmatrixni o'rnatish

1. `package` ma'lumotini ko'ring.
2. `apt` bilan `cmatrixni` o'rnatish.
3. `commandni` ishga tushirish; `Ctrl+C` bilan chiqing.
4. `package` versiyasini va `/usr/bin/cmatrixni` qaysi `package` berganini tekshiring.

### Tekshiruv

Ikki natija: A topshirig'i uchun bitta, B topshirig'i uchun bitta. Topshiriladigan `file` yo'q.

```
jdu-check M3
```

Foydali commandlar: `systemctl status`, `systemctl show`, `ps`, `kill`, `apt show`, `apt install`, `dpkg-query`, `command -v`, `dpkg -S`.

## M4 systemd service

### Maqsad

unit file, service va process munosabatini tekshirish; active va enabledni farqlash.

### Boshlang'ich holat

O'qituvchi tayyorlagan jdu-status.service loaded, inactive, disabled holatida. unit file tayyor.

### Topshiriq

1. systemctl cat jdu-status.service yordamida User, WorkingDirectory, ExecStartni o'qing.
2. unit fileni o'zgartirmay serviceni start qiling.
3. Tizim yoqilganda avtomatik start bo'ladigan qilib enable qiling.
4. active va enabled holatlarini alohida tekshiring.
5. Main PIDni topib, haqiqiy process bilan solishtiring.
6. processning useri, command linei va ish directorysi unitdagi qiymatlarga mosligini tekshiring.

### Tekshiruv

Uchta natija: active, enabled, o'zgarmagan unit bilan haqiqiy process mosligi. Topshiriladigan file yo'q.

```
jdu-check M4
```

Foydali: systemctl cat, systemctl start, systemctl enable, systemctl is-active, systemctl is-enabled, systemctl show, ps, /proc/PID/cmdline, /proc/PID/cwd.

## M5 Port, socket, and log

### Maqsad

service, process, socket, IP address, port, HTTP va journalni bitta ishlayotgan tizimning qismlari sifatida bog'lash.

### Boshlang'ich holat

O'qituvchi tayyorlagan jdu-web.service loaded, inactive, disabled holatida. unit file va kontent filei tayyor. TCP 8081da listener yo'q.

### Topshiriq

1. unit filedagi User, ExecStart, manzil, port va kontent pathini o'qing.
2. unit fileni o'zgartirmay jdu-web.serviceni start qiling.
3. 127.0.0.1:8081da aloqa kutayotgan socketni tekshiring. Bu holat listener (listening socket) deyiladi.

```
sudo ss -ltnp | grep ':8081'
```

1. ssning users: qatoridagi PID bilan service Main PIDini solishtiring.

```
systemctl show --property MainPID --value jdu-web.service
```

1. 0.0.0.0:8081 yoki [::]:8081da kutmayotganini tekshiring.
2. http://127.0.0.1:8081/ga request yuborib, HTTP responseni tekshiring.
3. service user jduweb kontent fileini o'qiy olishini tekshiring. jduweb login uchun emas; su - jduweb ishlatmang. Haqiqiy o'qish huquqi quyidagicha tekshiriladi: 0 — mumkin, 1 — mumkin emas.

```
sudo -u jduweb -- test -r /srv/jdu-web/index.txt
echo $?
```

user va groupni getent passwd jduweb hamda id jduweb bilan ko'ring. file va yuqori directorylarning permissionini stat va namei -l /srv/jdu-web/index.txt bilan tekshiring.

1. http://127.0.0.1:8081/m5-checkga request yuboring.
2. Hozirgi service ishga tushgan davrga tegishli jurnalda REQUEST path=/m5-checkni toping.
3. TCP 18081da listener yo'qligini va HTTP request muvaffaqiyatsiz bo'lishini tekshiring.

### Tekshiruv

To'rtta natija: listener va PID; HTTP va file permission; talaba yuborgan request logi; ochiq va yopiq portni solishtirish. LabCheckning / requesti talabani /m5-check requesti o'rnini bosmaydi.

```
jdu-check M5
```

Foydali: systemctl cat, systemctl start, systemctl show, sudo ss -lntp, curl, sudo -u, sudo journalctl.

## M6 SSH and remote operation

### Maqsad

CloudShellni local, Ubuntuni remote deb farqlash; SSH remote command va scp bilan upload/download qilish.

### Boshlang'ich holat

SSH kaliti, jdu-ubuntu aliasi va Session Manager tunneli tayyor. Ular topshiriq balliga kirmaydi. CloudShell va Ubuntuda Mission filelari hali yo'q.

### Topshiriq

1. CloudShellda id -un, hostname, pwdni tekshiring.
2. ssh jdu-ubuntu yoki SSH remote command orqali Ubuntudagi shu uch qiymatni tekshiring.
3. CloudShelldagi ~/jdu-lab/m6/local-source.txtga faqat quyidagi bitta satrni yozing:

```
JDU SSH transfer test
```

1. scp bilan Ubuntudagi ~/jdu-lab/m6/upload.txtga upload qiling.
2. SSH remote command bilan Ubuntuda ~/jdu-lab/m6/remote-result.txt yarating. Qiymatlarni qo'lda yozmang.

```
REMOTE_USER=<Ubuntuda aniqlangan user>
REMOTE_HOST=<Ubuntuda aniqlangan hostname>
REMOTE_PATH=<Ubuntuda aniqlangan home directory>
```

1. `scp` bilan Ubuntudagi `remote-result.txt`ni CloudShelldagi `~/jdu-lab/m6/downloaded-result.txt`ga `download` qiling.
2. `upload` manbasi va manzili, `download` manbasi va manzilini `sha256sum` bilan solishtiring.

### Tekshiruv

Ubuntu tomonida ikki topshiriq baholanadi. SSH orqali Ubuntuga kirib bajaring. Natija Dashboarddagi M6 Ubuntuga yuboriladi.

```
jdu-check M6
```

CloudShell tomonida to'rt topshiriq baholanadi. `exit` bilan Ubuntudan CloudShellga qaytgach bajaring. Natija Dashboarddagi M6 CloudShellga yuboriladi.

```
jdu-check M6
```

Dashboard M6 qatorida Ubuntu 2/2, CloudShell 4/4, jami 6/6ni alohida ko'rsatadi. Ikkala tomonda ham hammasi `PASS` bo'lgandagina M6 tugaydi. Bir tomonni qayta tekshirish boshqa tomondagi oxirgi natijani o'chirmaydi.

Ikkala tomonni to'liq `PASS` qiling. `private key`ni ko'rsatmang, topshirmang, ko'chirmang. SSH `servicei`, SSH sozlamasi va `authorized_keys`ni o'zgartirmang.

Foydali: `ssh`, `scp`, `sha256sum`, `id`, `hostname`, `pwd`, `printf`.

## M7 Integrated Ubuntu check

### Maqsad

M2, M4, M5dagi bilimlarni yangi `service`ga mustaqil qo'llash. Yangi `command` kiritilmaydi.

### Boshlang'ich holat

- O'qituvchi bergan `jdu-final.service` `loaded`, `inactive`, `disabled` holatida.
- `/srv/jdu-final` va `index.txt` hali yo'q.
- `jduwriter` va `jduviewer` `finalops` a'zosi emas.

### Topshiriq

1. `systemctl cat jdu-final.service` orqali `User`, `WorkingDirectory`, `ExecStart`ni o'qing.
2. `/srv/jdu-final`ni `root:finalops`, `mode 2775` qilib yarating.
3. `/srv/jdu-final/index.txt` yarating. Mazmuni `/etc/jdu-lab/final-marker`dagi bir satr bilan aynan bir xil bo'lsin.
4. `index.txt`ni `root:finalops`, `mode 664` qiling.
5. `jduwriter`ni `finalops`ga qo'shing. `jduviewer`ni qo'shmang.
6. `writer` kontentni o'zgartira olishini, `viewer` esa o'zgartira olmasligini tekshiring.
7. `unit` fileni o'zgartirmay `serviceni start` va `enable` qiling.
8. `service jdufinal` nomidan ishlashi va faqat `127.0.0.1:8090`da kutishini tekshiring.
9. `HTTP response status 200` bo'lib, kerakli `markerni` qaytarishini tekshiring.

10. `http://127.0.0.1:8090/m7-checkga request` yuboring.

11. Hozirgi `service` ishga tushgan davrga tegishli `journal`dan `REQUEST path=/m7-check`ni toping.

`user`ni almashtirish kerak bo'lsa, `ssm-user`dan `sudo su - jduwriter` yoki `sudo su - jduviewer`ni bajaring. Har tekshiruvdan keyin `exit` bilan `ssm-user`ga qayting. Oxirida `id -unni` tekshirib, keyin `jdu-check M7`ni bajaring.

### Tekshiruv

Oltita natija: `filelar` tuzilishi, `writer/viewer` huquqlari, `service`, `socket`, `HTTP` va `journal`. `unit file`, `SSH` sozlamasi va `LabCheck`ni o'zgartirmang.

```
jdu-check M7
```

Foydali: `id`, `getent`, `stat`, `chown`, `chmod`, `systemctl`, `ps`, `sudo ss -lntp`, `curl`, `sudo journalctl`.

## Xavfsizlik uchun taqiqlangan amallar

- `chmod 777`ni ishlatmang.
- `/etc/ssh/sshd_config`ni o'zgartirmang.
- `ssh.serviceni` to'xtatmang.
- `killall` yoki keng qamrovli `pkill`ni ishlatmang.
- `private key`ni ko'rsatmang, topshirmang va `Gitga` qo'shmang.
- Tashqi `hostlarni scan` qilmang yoki ularga `login` qilishga urinmang.